Assignment 1

Deadline: Monday, 19 October 2026, 18:00

Introduction

In this assignment, you will estimate a probability distribution from observations, answer probabilistic queries, investigate independence assumptions, and derive a factorised probabilistic model.

The file `student.csv` contains observations of five discrete random variables[1] and is available on Moodle:

Variable	Name	Values
D	Difficulty	easy (d^0), hard (d^1)
I	Interest	low (i^0), high (i^1)
G	Grade	A (g^1), B (g^2), C (g^3)
S	SAT	low (s^0), high (s^1)
L	Letter	weak (l^0), strong (l^1)

Use your preferred programming language and packages to solve the following tasks. We recommend using Python and representing the full joint distribution as a `numpy` array for this assignment.

Submission. Submit your solution via Moodle:

- Submit a PDF slide deck with up to four slides, one for each regular task (Tasks 1–4).
- You may add one additional slide for each bonus task you complete (Tasks 5–6).
- "Expected content on the solution slide" gives instructions on what to put on your slides.
- Additionally, submit a Jupyter notebook (preferred) or a similar executable script that makes your experiments reproducible. If your code consists of multiple files, submit them together as a ZIP archive.

Do not forget to:

- indicate which parts of the assignment you have completed here
- indicate which bonus assignments you have solved here;
- select an interview time slot here;
- proivde feedback here.

[1] The full example will be covered in the next lecture. For a preview, see Figure 3.3 on page 52 of *Probabilistic Graphical Models* by Koller and Friedman.

1 Empirical full joint distribution (1 point)

Construct the empirical joint distribution

$$\widehat{P}_{\text{emp}}(D, I, G, S, L)$$

from the data set `student.csv` using you preferred programming language.

Expected content on the solution slide: Explain how you represent and construct the full joint distribution. Report the shape and number of non-redundant parameters. You do not need to display the complete table on the slide.

2 Probabilistic queries (1 point)

Use $\widehat{P}_{\text{emp}}$ and answer the following probabilistic queries via inference by enumeration:

- What is the probability that a student receives a strong recommendation letter?
- Given a high SAT score, what is the probability that the student has high interest?
- Given an A grade, what is the probability that the course was hard?
- Given a strong recommendation letter and a high SAT score, what is the probability that the student has high interest?

Expected content on the solution slide: For each query, write the corresponding probabilistic query in mathematical notation. Report the results for all queries. For the last or the second to last query, write down the intermediate steps of inference by enumeration in mathematical notation.

3 Independence in the empirical distribution (1 point)

Investigate the following independence claims:

$$D \perp I,$$

$$S \perp \{D, G\} \mid I,$$

$$L \perp \{D, I, S\} \mid G.$$

Expected content on the solution slide: Report your results for all claims. Briefly explain how you investigated the last or second to last independence claim.

4 Factorisation and graphical model (1 point)

For the remainder of the assignment, assume that the three independence claims above hold exactly.

Starting from the chain rule with ordering

$$D, I, G, S, L,$$

derive a factorisation of

$$P(D, I, G, S, L)$$

in which each conditional distribution contains only those conditioning variables that cannot be eliminated using the stated independence assumptions.

How many independent numerical parameters are required by the full joint distribution, and how many by the factorised model?

Expected content on the solution slide: State all relevant steps in the derivation. State shape and numer of non-redundant parameters for each CPD. Draw the directed graph corresponding to the resulting factorisation. Use one node for each random variable and draw a directed edge $X \to Y$ whenever X occurs in the conditioning set of the factor for Y.

5 Bonus: Model-based full joint distribution (1 point)

Estimate all local probability distributions required by your factorisation using maximum-likelihood estimates (or something similar) from the data. Then construct the full joint distribution

$$\widehat{P}_{\text{model}}(D, I, G, S, L)$$

using the factorisation and the estimated CPDs.

Expected content on the solution slide: Explain how you calculated the CPDs and present selected tables (if there is enough space). If a required parent configuration does not occur in the data, explain how you deal with this case and justify your choice. Explain how you construct the full joint distribution from the CPDs.

6 Bonus: Compare the two distributions (1 point)

Compare

$$\widehat{P}_{\text{emp}} \qquad \text{and} \qquad \widehat{P}_{\text{model}}.$$

Expected content on the solution slide: Briefly answer the following questions:

1. Why are the two joint distributions not necessarily identical, even though they were estimated from the same data?
2. Recompute the independence and conditional-independence analyses from Task 3 using $\widehat{P}_{\text{model}}$. What changes?